AP CALCULUS AB

Unit 1 • Worksheet 1-4A • Answer key

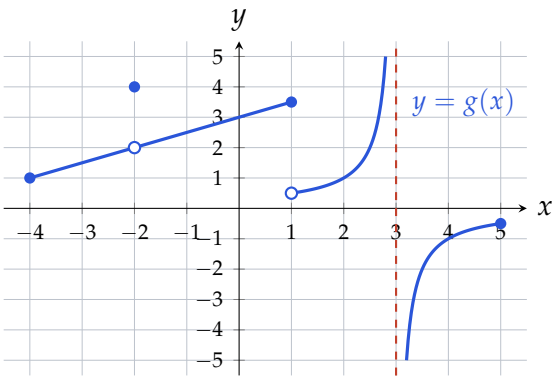
Worksheet 1-4A • Continuity at a Point

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Questions 1 to 4 refer to the graph of g on $[-4, 5]$.



Question 1

NO CALCULATOR

What type of discontinuity does g have at $x = -2$?

(A) Removable (hole)
(B) Jump
(C) Infinite
(D) None

Solution

Answer: (A). The limit exists (2) but $g(-2) = 4$.

Question 2

NO CALCULATOR

What type of discontinuity does g have at $x = 1$?

(A) Removable
(B) Jump
(C) Infinite
(D) None

Solution

Answer: (B). Left limit 3.5, right limit 0.5.

Question 3

NO CALCULATOR

What type of discontinuity does g have at $x = 3$?

(A) Removable
(B) Jump
(C) Infinite
(D) None

Solution
Answer: (C). A vertical asymptote: $g \to +\infty$ from the left and $-\infty$ from the right.

Question 4

NO CALCULATOR

At $x = -2$, which part of the continuity test fails?

(A) The left limit does not equal the right limit.
(B) $g(-2)$ is not defined.
(C) The limit is infinite.
(D) The limit does not equal $g(-2)$.

Solution
Answer: (D). Left $=$ right $= 2$, but $g(-2) = 4$.

Question 5

NO CALCULATOR

Let $h(x) = \begin{cases} x^2 + 2, & x \leq 1 \\ 4x - 1, & x > 1. \end{cases}$ Which is true at $x = 1$?

(A) h is continuous: $3 = 3 = 3$.
(B) Jump discontinuity.
(C) Removable discontinuity.
(D) $h(1)$ is undefined.

Solution
Answer: (A). Left 3, right 3, $h(1) = 3$.

Question 6

NO CALCULATOR

For what value of k is $g(x) = \begin{cases} kx - 1, & x < 2 \\ x^2 + k, & x \geq 2 \end{cases}$ continuous at $x = 2$?

(A) 3
(B) 5
(C) -5
(D) 1

Solution
Answer: (B). $2k - 1 = 4 + k$, so $k = 5$.

Question 7

NO CALCULATOR

For what value of k is $f(x) = \begin{cases} kx^2, & x \le -1 \\ 2x + k, & x > -1 \end{cases}$ continuous at $x = -1$?

(A) 1
(B) -1
(C) No value of k works.
(D) Every value of k works.

Solution
Answer: (C). $k = -2 + k$ gives $0 = -2$: impossible.

Question 8

NO CALCULATOR

Which function is continuous at $x = 0$?

(A) $\dfrac{x^2 - x}{x}$
(B) $\begin{cases} 1, & x < 0 \\ 0, & x \ge 0 \end{cases}$
(C) $\dfrac{1}{x^2}$
(D) $|x|$

Solution
Answer: (D). (A): hole at 0. (B): jump. (C): infinite.

Question 9 (Free response)

NO CALCULATOR

(a) Let $p(x) = \begin{cases} 3x - 2, & x \le 2 \\ x^2, & x > 2. \end{cases}$ Prove that p is continuous at $x = 2$ with the continuity test.

(b) Find k so that $q(x) = \begin{cases} x^2 + k, & x < 3 \\ kx + 1, & x \ge 3 \end{cases}$ is continuous at $x = 3$.

(c) Find and classify the discontinuities of $r(x) = \dfrac{x^2 - 1}{x^2 - x}$.

Solution
(a) Left $3(2) - 2 = 4$; right $2^2 = 4$; $p(2) = 4$. $4 = 4 = 4$: continuous.
(b) $9 + k = 3k + 1$, so $k = 4$.
(c) $r = \frac{(x-1)(x+1)}{x(x-1)}$: removable at $x = 1$ (limit 2), infinite at $x = 0$.